

Chaoran Chen

Curriculum Vitae

Department of Computer Science and Engineering
College of Engineering
University of Notre Dame
Notre Dame, IN 46556 USA

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Research Interests

Human-AI Interaction, Privacy and Security.

Education

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|---|-------------|
| Ph.D. in Computer Science and Engineering
University of Notre Dame, <i>Notre Dame, IN</i>
Department of Computer Science and Engineering
Advisors: Toby Jia-Jun Li and Yanfang Ye | 2022 – 2026 |
| M.S. in Educational Technology and Applied Learning Sciences
Carnegie Mellon University, <i>Pittsburgh, PA</i>
Human Computer Interaction Institute, School of Computer Science | 2021 – 2022 |
| B.E. in Industrial Design
Tongji University, <i>Shanghai, China</i>
College of Design and Innovation | 2015 – 2020 |

Selected Honors and Awards

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| Third prize of Tech Ethics Hackathon (\$1,000), Notre Dame-IBM Technology Ethics Lab | 2024 |
| Best Implementation Award (\$2,000), MobileHCI 2022 | 2022 |
| Merit Scholarship (\$7,000), Carnegie Mellon University | 2021 |

Major Refereed Conference Papers

(* indicate equal contribution)

- [C.13] **Dark Patterns Meet GUI Agents: LLM Agent Susceptibility to Manipulative Interfaces and the Role of Human Oversight**
Jingyu Tang*, **Chaoran Chen***, Jiawen Li, Zhiping Zhang, Bingcan Guo, Ibrahim Khalilov, Simret Araya Gebreegzabher, Bingsheng Yao, Dakuo Wang, Yanfang Ye, Tianshi Li, Ziang Xiao, Yaxing Yao, and Toby Jia-Jun Li
Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI 2026)
- [C.12] **Through the Lens of Human-Human Collaboration: A Configurable Research Platform for Exploring Human-Agent Collaboration**
Bingsheng Yao, Jiaju Chen, **Chaoran Chen**, April Wang, Toby Jia-jun Li, and Dakuo Wang
Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI 2026)

- [C.11] **My Favorite Streamer is an LLM: Discovering, Bonding, and Co-Creating in AI VTuber Fandom**
Jiayi Ye, **Chaoran Chen**, Yue Huang, Yanfang Ye, Yaxing Yao, and Toby Jia-Jun Li
Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI 2026)
- [C.10] **The Behavioral Fabric of LLM-Powered GUI Agents: Human Values and Interaction Outcomes**
Simret Araya Gebreegziabher, Yukun Yang, Charles Chiang, Hojun Yoo, **Chaoran Chen**, Hyo Jin Do, Zahra Ashktorab, Werner Geyer, Diego Gómez-Zarà, and Toby Jia-Jun Li
Proceedings of the 31th ACM Conference on Intelligent User Interfaces (IUI 2026)
- [C.9] **Why Am I Seeing This: Democratizing End User Auditing for Online Content Recommendations**
Chaoran Chen, Leyang Li, Luke Cao, Yanfang Ye, Tianshi Li, Yaxing Yao, and Toby Jia-Jun Li
Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology (UIST 2025)
- [C.8] **Towards a Design Guideline for RPA Evaluation: A Survey of Large Language Model-Based Role-Playing Agents**
Chaoran Chen, Bingsheng Yao, Ruishi Zou, Wenyue Hua, Weimin Lyu, Toby Jia-Jun Li, and Dakuo Wang
Findings of the Association for Computational Linguistics (ACL 2025 Findings)
- [C.7] **CLEAR: Towards Contextual LLM-Empowered Privacy Policy Analysis and Risk Generation for Large Language Model Applications**
Chaoran Chen, Daodao Zhou, Yanfang Ye, Toby Jia-jun Li, and Yaxing Yao
Proceedings of the 30th ACM Conference on Intelligent User Interfaces (IUI 2025)
- [C.6] **Careful About What App Promotion Ads Recommend! Detecting and Explaining Malware Promotion via App Promotion Graph**
Shang Ma*, **Chaoran Chen***, Shao Yang, Shifu Hou, Toby Jia-Jun Li, Xusheng Xiao, Tao Xie, and Yanfang Ye
The Network and Distributed System Security Symposium (NDSS 2025)
- [C.5] **Symbolic Prompt Tuning Completes the App Promotion Graph**
Zhongyu Ouyang, Chunhui Zhang, Shifu Hou, Shang Ma, **Chaoran Chen**, Toby Li, Xusheng Xiao, Chuxu Zhang, and Yanfang Ye
Machine Learning and Knowledge Discovery in Databases (ECML PKDD 2024)
- [C.4] **An Empathy-Based Sandbox Approach to Bridge the Privacy Gap among Attitudes, Goals, Knowledge, and Behaviors**
Chaoran Chen, Weijun Li, Wenxin Song, Yanfang Ye, Yaxing Yao, and Toby Jia-Jun Li
Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI 2024)
- [C.3] **ARDW: An augmented reality workbench for printed circuit board debugging**
Ishan Chatterjee, Tadeusz Pforte, Aspen Tng, Farshid Salemi Parizi, **Chaoran Chen** and Shwetak Patel
Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology (UIST 2022)
- [C.2] **Patterns for Representing Knowledge Graphs to Communicate Situational Knowledge of Service Robots**
Shengchen Zhang, Zixuan Wang, **Chaoran Chen**, Yi Dai, Lyumanshan Ye and Xiaohua Sun

Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI 2021)

[C.1] **AI-Sketcher: A Deep Generative Model for Producing High-Quality Sketches**

Nan Cao, Xin Yan, Yang Shi, and **Chaoran Chen**

Proceedings of the AAAI Conference on Artificial Intelligence (AAAI 2019)

Refereed Workshop Papers

[W.5] **Iterative Contextual Consent: AI-enabled Data Privacy Contracts**

Daniel D Slate, **Chaoran Chen**, Yaxing Yao, Toby Jia-Jun Li

CCS 2025 Workshop on Human-Centered AI Privacy and Security

[W.4] **Beyond Permissions: Investigating Mobile Personalization with Simulated Personas**

Ibrahim Khalilov, **Chaoran Chen**, Ziang Xiao, Tianshi Li, Toby Jia-Jun Li, Yaxing Yao

CCS 2025 Workshop on Human-Centered AI Privacy and Security

[W.3] **Toward a Human-centered Evaluation Framework for Trustworthy LLM-powered GUI Agents**

Chaoran Chen, Zhiping Zhang, Ibrahim Khalilov, Bingcan Guo, Simret A. Gebreegziabher, Yanfang Ye, Ziang Xiao, Yaxing Yao, Tianshi Li, and Toby Jia-Jun Li

CHI 2025 Workshop on Human-Centered Evaluation and Auditing of Language Models (HEAL)

[W.2] **Evaluating the LLM Agents for Simulating Humanoid Behavior**

Chaoran Chen, Bingsheng Yao, Yanfang Ye, Dakuo Wang, and Toby Jia-Jun Li

CHI 2024 Workshop on Human-Centered Evaluation and Auditing of Language Models (HEAL)

[W.1] **Prompt Learning Unlocked for App Promotion in the Wild**

Zhongyu Ouyang, Shifu Hou, Shang Ma, **Chaoran Chen**, Chunhui Zhang, Toby Jia-Jun Li, Xusheng Xiao, Chuxu Zhang, and Yanfang Ye

NeurIPS 2023 Workshop on New Frontiers in Graph Learning (GLFrontiers)

Relevant Research Experience

Applied Science Intern

Apr. 2026-Jul. 2026

Customer Service Chatbot Team

Amazon, Seattle, WA, USA

Host: Dr. Zhou Yu

Student Researcher

Jan. 2026-Apr. 2026

Workspace & Deepmind

Google, Sunnyvale, CA, USA

Host: Dr. Dayu Yuan and Dr. Peter Kairouz

Student Researcher

Jul. 2025-Nov. 2025

Workspace & Deepmind

Google, Sunnyvale, CA, USA

Host: Dr. Dayu Yuan and Dr. Peter Kairouz

Research Intern

Oct. 2024-Dec. 2024

Technology Harm Prevention Group

Max Planck Institute for Security and Privacy, Bochum, Germany
Host: Dr. Abraham Mhaidli

Teaching Experience

Teaching Assistant , <i>CSE 40567: Computer Security</i> Department of Computer Science and Engineering, University of Notre Dame	Spring 2025
Teaching Assistant , <i>CSE 40625: Machine Learning</i> Department of Computer Science and Engineering, University of Notre Dame	Fall 2023
Teaching Assistant , <i>05-738: Evidence-Based Educational Design</i> Human-Computer Interaction Institute, Carnegie Mellon University	Fall 2022
Teaching Assistant , <i>05-440/05-640: Interaction Techniques (IxT)</i> Human-Computer Interaction Institute, Carnegie Mellon University	Spring 2022

Professional Service Academic Service

Academic Service

Program Committee Member

Human-Agent Collaboration Workshop@ACM CHI(2026), HAIPS workshop@ACM CCS (2025)

Conference Reviewer

ACM CHI (2023-2026), ACM CSCW (2025), ACM IUI (2025-2026)

Received “special recognitions” for outstanding reviews in ACM CHI 2024 Late-Breaking Work, ACM CSCW 2025, ACM IUI 2025-2026, and ACM CHI 2026.

Journal Reviewer

ACM TOPS (2022, 2025)

Departmental and Community Service

Student Volunteer, ACM MobileHCI 2022